PRACTICAL NO 2. Describing the Distribution of a variable.

A)Aim:- To calculate descriptive statistics for a business dataset.

Dataset:- Sales.CSV

	Product	Category	Sales
1	Laptop	Electronics	78000
2	Mobile	Electronics	40000
3	Chair	Furniture	10000
4	Table	Furniture	15000
5	Printer	Electronics	35000
6	Sofa	Furniture	45000
7	AC	Electronics	96000
8	Fan	Electronics	25000
9	Cupboard	Furniture	30000
10	TV	Electronics	88000
11	Bed	Furniture	56000
12	Washing Machine	Electronics	76000
13	Refrigerator	Electronics	87000
14	Desk	Electronics	20000
15	Bookshelf	Furniture	5000

Code:-

#Import Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

#Load dataset

```
df=pd.read_csv("sales2.csv")
```

#Printing dataset

```
print("dataset")
print(df)
```

```
Dataset
            Product     Category  Sales
0            Laptop  Electronics  78000
1            Mobile  Electronics  40000
2             Chair    Furniture  10000
3             Table    Furniture  15000
4           Printer  Electronics  35000
5              Sofa    Furniture  45000
6                AC  Electronics  96000
7               Fan  Electronics  25000
8         Cupboard    Furniture  30000
9                TV  Electronics  88000
10              Bed    Furniture  56000
11  Washing Machine  Electronics  76000
12     Refrigerator  Electronics  87000
13             Desk  Electronics  20000
14        Bookshelf    Furniture   5000
```

Descriptive Statistics

```
print("\nDescroptive Statistics")
print(df["Sales"].describe())
```

```
Descriptive Statistics
count       15.000000
mean     47066.666667
std      30961.881326
min       5000.000000
25%      22500.000000
50%      40000.000000
75%      77000.000000
max      96000.000000
Name: Sales, dtype: float64
```

Mean, Median, Mode

```
print("\nMean:",df["Sales"].mean())
print("\nMedian:",df["Sales"].median())
print("\nMode:",df["Sales"].mode())
```

```
Mean: 47066.666666666664

Median: 40000.0

Mode: 0      5000
1     10000
2     15000
3     20000
4     25000
5     30000
6     35000
7     40000
8     45000
9     56000
10    76000
11    78000
12    87000
13    88000
14    96000
Name: Sales, dtype: int64
```

#Variance, Standard Deviation

```
print("\nVariance:",df["Sales"].var())
print("\nStandard Deviation:",df["Sales"].std())
```

```
Variance: 958638095.2380952

Standard Deviation: 30961.88132588353
```

#Minimum, Maximum

```
print("\nMinimum:",df["Sales"].min())
print("\nMaximum:",df["Sales"].max())
```

```
Minimum: 5000

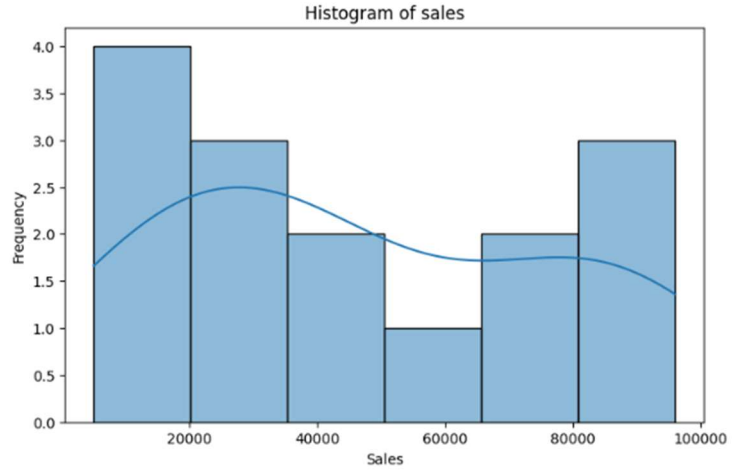
Maximum: 96000
```

B)Aim:- Visualize the distribution of variable using histogram and box plot

Code:- (Histogram)

```
plt.figure(figsize=(8,5))
sns.histplot(df["Sales"],bins=6,kde=True)
plt.title("Histogram of sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
```

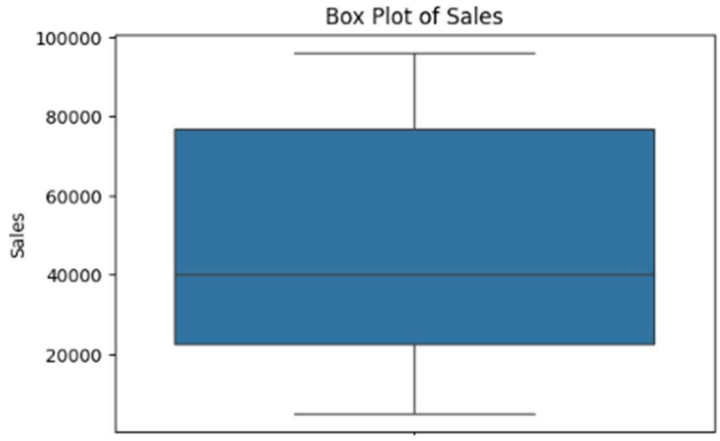
Output:-



Code:- (Box Plot)

```
plt.figure(figsize=(6,4))
sns.boxplot(y=df["Sales"])
plt.title("Box Plot of Sales")
plt.show()
```

Output:-



Conclusion:-

The sales dataset was successfully analyzed using various descriptive statistical methods. Mean, median, mode, variance, standard deviation, minimum, and maximum values were calculated to understand the sales data in detail. The distribution of sales was also represented visually using a histogram and a box plot, which helped in understanding the frequency, spread, and variation of the data. Overall, this practical provided a clear understanding of how descriptive statistics and graphical methods can be used to analyze and interpret a business dataset.